Funciones especiales | Clase # 10 | 26/08/2025

Laplaciano

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = x^2 - y^2$$

Cambiar de variable

$$\left.\begin{aligned} z &= x + iy \\ \bar{z} &= x - iy \end{aligned}\right\} \qquad x = \frac{z + \bar{z}}{2} \qquad y = \frac{z - \bar{z}}{2i}$$

Si tenemos una función compleja de la forma:

$$f(z) = u(x,y) + i\,v(x,y)$$

Analítica $\longrightarrow$ Cauchy–Riemann

$$\lim_{\substack{\Delta x \to 0 \\ \Delta y \to 0}} \frac{f(z+\Delta z) - f(z)}{\Delta z}$$

$$\lim_{\substack{\Delta x \to 0 \\ \Delta y \to 0}} \left\{ \frac{u(x+\Delta x, y+\Delta y) + i v(x+\Delta x, y+\Delta y) - u(x,y) - i v(x,y)}{\Delta x + i \Delta y} \right\}$$

Tenemos

$$\lim_{\Delta y \to 0} \left\{ \frac{u(x+\Delta x, y) + iv(x+\Delta x, y) - u(x,y) - iv(x,y)}{\Delta x} \right\}$$

$i\Delta y$

$$\frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$$

Tenemos

$$- i \frac{\partial u}{\partial y} + \frac{\partial v}{\partial y}$$

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \; ; \; - \frac{\partial v}{\partial x} = \frac{\partial u}{\partial y}$$

Armónicas

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 v}{\partial x \partial y}$$

$$\Rightarrow \quad \frac{\partial^2 u}{\partial y^2} = - \frac{\partial^2 v}{\partial y \partial x}$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}$$

si $f(x,y)$ es una función real:

$$f(x,y) \longrightarrow f\left(\frac{z+\bar{z}}{2}, \frac{z-\bar{z}}{2i}\right) = G(z,\bar{z})$$

si $A(x,y) = P(x,y) + i\,Q(x,y)$

Regla de la cadena

Tenemos

$$\frac{\partial F(z,\bar{z})}{\partial x} = \frac{\partial z}{\partial x}\frac{\partial F}{\partial z} + \frac{\partial \bar{z}}{\partial x}\frac{\partial F}{\partial \bar{z}}$$

$$\vdots$$

$$= \frac{\partial F}{\partial z} + \frac{\partial F}{\partial \bar{z}}$$

$$\frac{\partial}{\partial x} \longrightarrow \frac{\partial}{\partial z} + \frac{\partial}{\partial \bar{z}}$$

$$\frac{\partial F(z,\bar{z})}{\partial y} = \frac{\partial z}{\partial y}\frac{\partial F}{\partial z} + \frac{\partial \bar{z}}{\partial y}\frac{\partial F}{\partial \bar{z}}$$

$$\vdots$$

$$= i\frac{\partial F}{\partial z} - i\frac{\partial F}{\partial \bar{z}}$$

$$\frac{\partial}{\partial y} \longrightarrow i\left(\frac{\partial}{\partial z} - \frac{\partial}{\partial \bar{z}}\right)$$

Operador nabla

$$\nabla = \frac{\partial}{\partial x} + i\frac{\partial}{\partial y} = \cancel{\frac{\partial}{\partial z}} + \frac{\partial}{\partial \bar{z}} - \cancel{\frac{\partial}{\partial z}} + \frac{\partial}{\partial \bar{z}} = 2\frac{\partial}{\partial \bar{z}}$$

$$\bar{\nabla} = \frac{\partial}{\partial x} - i\frac{\partial}{\partial y} = 2\frac{\partial}{\partial z}$$

$$\nabla A = \frac{\partial A}{\partial x} + i\frac{\partial A}{\partial y} = 2\frac{\partial A}{\partial \bar{z}}$$

$$\nabla \cdot A \longrightarrow Re\{\bar{\nabla} A\} = Re\left\{\left(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\right)(P + iQ)\right\}$$

$$\vdots$$

$$= \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} = Re\ 2\frac{\partial}{\partial}$$

Rotacional

$$\nabla \times A \longrightarrow \left(0, 0, \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)$$

$$|\nabla \times A| = \left|Im\{\bar{\nabla} A\}\right| = \left|Im\left(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\right)(P + iQ)\right.$$

$$\vdots$$

$$= \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = \operatorname{Im} 2 \frac{\partial A}{\partial z}$$

$$4 \frac{\partial^2 u}{\partial z \partial \bar{z}} = \left(\frac{z+\bar{z}}{2}\right)^2 - \left(\frac{z-\bar{z}}{2i}\right)^2$$

$$\vdots$$

$$= \frac{1}{4}\left(z^2 + 2z\bar{z} + \bar{z}^2\right) + \frac{1}{4}\left(z^2 - 2z\bar{z} + \bar{z}^2\right)$$

$$\frac{\partial^2 u}{\partial z \partial \bar{z}} = \frac{1}{8}\left(z^2 + \bar{z}^2\right)$$

$$\frac{\partial u}{\partial z} = \frac{1}{8}\left[\frac{z^3}{3} + \bar{z}^2 z\right] + F(\bar{z})$$

$$u = \frac{1}{8}\left[\frac{z^3}{3}\bar{z} + \frac{\bar{z}^3}{3} z\right] + \int F(\bar{z})\, d\bar{z} + G(z)$$

Ecuación de Laplace:

$$\nabla^2 \phi(x,y,z) = 0$$

- cartesianas rectangulares
- cilíndrica
- Esférica

Ecuación de Hemholtz:

$$\left(\nabla^2 \phi + k^2\right)\phi = 0$$

Cartesianas

Separación de variables:

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} = 0$$

$$\Phi(x,y,z) = X(x)Y(y)Z(z)$$

$$YZ\frac{d^2X}{dx^2} + XZ\frac{d^2Y}{dy^2} + XY\frac{d^2Z}{dz^2} = 0$$

Tenemos

$$\underbrace{\frac{1}{X}\frac{d^2X}{dt^2}}_{-m^2} + \underbrace{\frac{1}{Y}\frac{d^2Y}{dy^2}}_{-n^2} + \underbrace{\frac{1}{Z}\frac{d^2Z}{dz^2}}_{\ell^2} = 0$$

Tenemos

$$X(x) = e^{imx}$$

$$Y(y) = e^{iny}$$

$$Z_1(z) = e^{\pm\sqrt{n^2+m^2}\,z}$$

Tenemos

$$\phi(x,y,z) = \sum_{n,m} C_{nm}\, e^{imx}\, e^{imy}\, e^{\pm\sqrt{n^2+m^2}}$$

Sea $u_1, \ldots, u_n(x)$ def. $[a,b]$

$$\int_a^b u_n(x)\, u_m(x)\, dx = \delta_{mn} S_n$$

$$S_n = 1$$

Tenemos

$$u_n'(x) = \frac{u_n(x)}{\int_a^b u_n(x)\, u_m^*(x)\, dx}$$

$$\vec{a} = \sum_{n=1} a_n u_n(x)$$

Tenemos

$$f(x) = -\sum_n a_n u_n(x) = 0$$

Entonces

$$M = \int_a^b \left| f(x) - \sum_n a_n u_n(x) \right|^2 dx$$

$\vdots$

$$\vdots$$

$$= \int_a^b \left(f(x) - \sum_n a_n u_n(x)\right)\left(f^*(x) - \sum_m a_m^* u_m^*(x)\right) dx$$

$$\vdots$$

$$= \int_a^b \Big\{ f(x) f^*(x) - \sum_m f(x)\, a_n^* u_n^*(x) - \sum_n f^*(x) a_n u_n(x)$$

$$+ \sum_n \sum_m a_n a_m^* \, u_n(x) u_m^*(x) \Big\} dx$$

Tenemos

$$\int_a^b a_m^* u_n(x) u_m^*(x)\, dx$$

$$a_m^* \int_a^b \underbrace{u_n(x)\, u_m^*(x)}_{\delta_{nm}}\, dx$$

Tenemos

$$\frac{\partial M}{\partial a_n} = 0 = a_n^* - \int_a^b f^*(x)\, u_n(x)\, dx = 0$$

$$\frac{\partial M}{\partial a_n^*} = a_n - \int_a^b f(x)\, u_n^*(x)\, dx = 0$$

$$a_n = \int_a^b f(x)\, u_n(x)\, dx$$

<u>Tenemos</u>

$$f(x) = \sum_n a_n\, u_n(x); \qquad u_n(x)\ ;\ [a,b]$$

$$\int_{-\pi}^{\pi} \operatorname{sen}(nx')\operatorname{sen}(mx')\, dx'\, dx' = \delta_{nm}\pi$$

$$\int_{-\pi}^{\pi} \frac{\operatorname{sen}(nx')}{\sqrt{\pi}}\, \frac{\operatorname{sen}(mx')}{\sqrt{\pi}}\, dx' = \delta_{nm}$$

$$\int_{-\pi}^{\pi} u_n(x')\, u_m^*(x')\, dx' = \delta_{nm}$$

$$f(x) = \sum_n a_n u_n(x)$$

$$\int_{-\pi}^{\pi} \frac{\cos(nx')}{\sqrt{\pi}} \frac{\cos(mx')}{\sqrt{\pi}} dx' = \delta_{nm}$$

$$\int_{-\pi}^{\pi} V_n(x') V_n^*(x') dx' = \delta_{nm}$$

$$f(x) = \sum_n a_n u_n(x) + \sum_n b_n V_n(x)$$

$$a_n = \int_{-\pi}^{\pi} f(x) u_n^*(x) dx$$

$$b_n = \int_{-\pi}^{\pi} f(x) V_n^*(x) dx$$

Análisis de Fourier $\rightarrow$ construcción

Condición de ortogonalidad:

$$\int u_n(x) u_m^*(x) dx = \delta_{nm}$$

Funciones de peso $\}$ conjunto completo (seno y coseno)

$$\int \rho(x)\, u_n(x)\, u_m^*(x)\, dx = \delta_{nm}$$

Problema de Sturm – Louville → Formas autoadjuntas

Introducimos el cambio de variable:

$$x' = \frac{2\pi}{d} x \quad ; \quad x = \frac{d}{2\pi} x'$$

$$\Rightarrow \int_{-\frac{d}{2}}^{\frac{d}{2}} \frac{1}{\pi} \operatorname{sen}\left(\frac{2\pi}{d} n x\right) \operatorname{sen}\left(\frac{2\pi m x}{d}\right) \frac{2\pi}{d}\, dx$$

$$\int_{-d/2}^{d/2} \underbrace{\sqrt{\frac{2}{d}} \operatorname{sen}\left(\frac{2\pi n x}{d}\right)}_{u_n(x) = \sqrt{\frac{2}{d}} \operatorname{sen}\left(\frac{2\pi n}{d} x\right)} \sqrt{\frac{2}{d}} \operatorname{sen}\left(\frac{2\pi m}{d} x\right) dx$$

ahora entre: $\left[-\frac{d}{2}, \frac{d}{2}\right]$

$$u_n(x) = \sqrt{\frac{2}{d}} \operatorname{sen}\left(\frac{2\pi n}{d} x\right)$$

$$V_n(x) = \sqrt{\frac{2}{d}} \cos\left(\frac{2\pi n}{d} x\right)$$

$$\int_{-d/2}^{d/2} u_n(x)\, u_m^*(x)\, dx = \delta_{nm}$$

$$\int_{-d/2}^{d/2} u_n(x)\, V_m(x)\, dx = 0$$

$$\int_{-d/2}^{d/2} V_n(x)\, V_m^*(x)\, dx = \delta_{nm}$$

$$f(x) = \sum_n a_n\, u_n(x) + \sum_n b_n\, V_n(x)$$

$$a_n = \int_{-d/2}^{d/2} f(x)\, u_n^*(x)\, dx$$

$$V_n(x) = \sqrt{\frac{2}{d}} \cos\left(\frac{2\pi n}{d} x\right) = \sqrt{\frac{2}{d}} \left\{ \frac{e^{i\frac{2\pi n}{d}x} + e^{-i\frac{2\pi}{d}n x}}{2} \right\}$$

$$u_n(x) = \sqrt{\frac{2}{d}} \sin\left(\frac{2\pi n}{d} x\right) = \sqrt{\frac{2}{d}} \left\{ \frac{e^{i\frac{2\pi n}{d}x} - e^{-i\frac{2\pi n x}{d}}}{2i} \right\}$$

$$b_n = \int_{-d/2}^{d/2} f(x)\, V_n^*(x)\, dx$$

$$V_n(x) = \frac{g_n(x) + g_{-n}(x)}{\sqrt{2}}$$

$$u_n(x) = \frac{g_n(x) - g_{-n}(x)}{\sqrt{2}\, i}$$

$$g_n(x) = \frac{1}{\sqrt{d}} e^{i \frac{2\pi n}{d} x}$$

$$g_{-n}(x) = \frac{1}{\sqrt{d}} e^{-i \frac{2\pi n}{d} x}$$

$$f(x) = \sum_{n=-\infty}^{\infty} a_n g_n(x)$$